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Bihar Engineering University, Patna

B.Tech 2nd Semester Exam., 2016

Subject: ELEMENTS OF MECHANICAL ENGINEERING

Code: 021201

Time: 3 hours

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer the following/Choose the correct option (any seven): **[2×7=14]**

(a) Give two examples of secondary energy sources.

(b) Which quality of coal is most suitable for thermal power plant?

(i) Peat (iii) Bituminous
(ii) Lignite (iv) Anthracite

(c) Ice kept in a wall insulated thermos flask is an example of which system?

(i) Closed system (iii) Open system
(ii) Isolated system (iv) Non-flow adiabatic system

(d) The long tube that carries water from the dam to the turbine is called

(i) spillway (iii) surge tank
(ii) draft tube (iv) penstock

(e) The processes of generation of steam in the pressure cooker is an example of

(i) constant pressure process (iii) constant enthalpy process
(ii) constant volume process (iv) constant entropy process

(f) In a boiler, the air preheater is invariably located between

(i) furnace and economiser (iii) economiser and feed pump
(ii) economiser and chimney (iv) furnace and feed pump

(g) In an aqua-ammonia absorption refrigeration system, the ammonia vapours are produced at high pressure in the

(i) absorber (iii) evaporator
(ii) condenser (iv) generator

(h) The component which maintains a constant angular motion of the crank within a cycle is called

(i) flywheel (iii) crankshaft
(ii) governor (iv) connecting rod

(i) In a four-stroke cycle diesel engine, during suction stroke

(i) mixture of fuel and air is sucked in (iii) only air is sucked in
(ii) only fuel is sucked in (iv) None of the above

(j) The property of material to resist fracture under impact loading is called

(i) strength (iii) brittleness
(ii) hardness (iv) toughness

Q.2 Solve both questions : **[14]**

(a) What are the advantages and disadvantages of renewable sources of energy?

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(b) What are the applications of solar energy? Discuss the working of a solar plate collector with the help of a suitable neat sketch.

Q.3 Solve both questions : **[14]**

(a) Explain the thermodynamic equilibrium.

(b) Air initially at a pressure of 75 kPa and temperature of 1000 K and occupying a volume of 0·12 m^3 is compressed isothermally until the volume is halved. Subsequently, the volume undergoes further compression at constant pressure till it is halved again. Calculate the work done.

Q.4 Solve all parts : **[14]**

(a) Define a boiler as per IBR. What are the purposes of steam generation?

(b) What is natural and forced circulation boiler?

(c) Explain the working of water level indicator with suitable diagram.

Q.5 Solve all parts : **[14]**

(a) Explain the working principle of reaction turbine.

(b) State the merits of gas turbine over IC engine.

(c) What are the applications of gas turbine?

Q.6 Solve both questions : **[14]**

(a) How do you classify IC engines? Represent Otto and Diesel cycle on $p-v$ and $T-S$ plot.

(b) Derive an expression for air standard efficiency of Otto cycle.

Q.7 Solve both questions : **[14]**

(a) Give the complete layout of steam power plant showing the appropriate position of economiser, air preheater, cooling tower and feed pump.

(b) What are the advantages and disadvantages of hydroelectric power plant?

Q.8 Solve all parts : **[14]**

(a) What is the principle of refrigeration?

(b) Compare the vapour absorption refrigeration system with vapour compression refrigeration system.

(c) Define air-conditioning.

Q.9 Solve both questions : **[14]**

(a) What is heat treatment of metals? What is quenching?

(b) What are the classifications of plain carbon steel? What are their properties and applications?

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